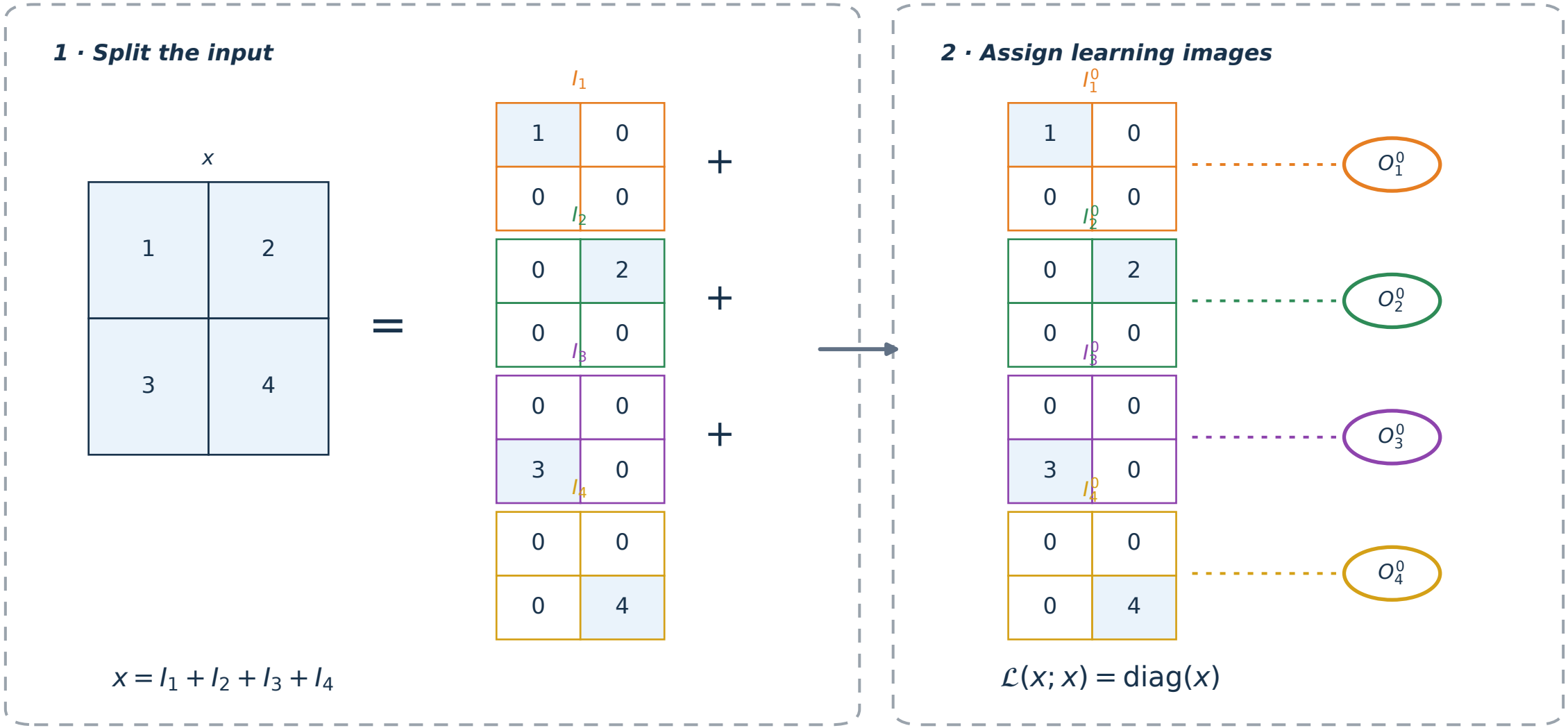Initialization: split the input, then assign one image to each coordinate